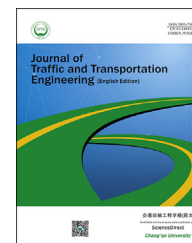


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Review Article

An overview of biofuel power generation on policies and finance environment, applied biofuels, device and performance



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HIGHLIGHTS

- Biomass, biogas and biofuel power generation are reviewed.
- Policies and finance environment of biofuel power generation are analyzed.
- Installations of biofuel power generation are analyzed.
- Sources and application of biofuels on power generation are analyzed.
- Performance and emissions of biofuel power generation are analyzed.

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ABSTRACT

The prices of fossil fuels, the gap between energy supply and demand, energy security, environmental issues, and the reduction of greenhouse gas emissions commonly promote the development of biomass power generation in China. Compared to direct biomass combustion power generation and biogas power generation, liquid biofuel power generation has better application potential. Policies and finance environment of countries and organizations have promoted the extensive application of biofuel power generation in recent years, especially in China. Evaluation of internal combustion engine, gas turbine and fuel cell on the stationary power station and mobile generation shows that gas turbine is the main generator-set, the fuel cell is the fastest-growing and future generator-set, while the engine is the first choice in transport and low power generation field. As the second, third and fourth generation biofuels, biomethanol, bioethanol, biodiesel and others have been researched and widely used in power generation. In addition, power generation performance and emissions have been improved to some extent after biofuels supplying. For the future, combined heat and power (CHP) generation and integrated gasification combined cycle (IGCC) is imperative development trends of power generation for higher

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energy and exergy efficiency. CHP based on the various generating facilities will be the next choice and direction of energy use trends for transport.

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1. Introduction

Electricity is 3.2 times more efficient than oil and 17.3 times more efficient than coal, that is, 1 ton of standard coal equivalent electricity creates the same economic value as 3.2 tons of standard coal equivalent oil and 17.3 tons of standard coal equivalent coal (Niu et al., 2017). In 2019, China's coal-fired power stations accounted for 50.2% of the world's coal-fired power generation (Chen et al., 2020; He et al., 2018). The development of coal and petroleum is limited, thus renewable sources will be the main trend of future energy, and it is necessary to develop large-scale development, especially in power generation (European Parliament, 2009). Global renewable energy resources are divided into solar energy, hydro energy, wind energy, biomass energy, etc. Among them, solar energy, hydro energy and wind energy are mainly used for power generation, but hydropower, wind power and solar photovoltaic power generation have strong volatility, intermittence and uncertainty, which has brought serious impact on the stable operation of the power grid (Wang et al., 2017). Solar thermal power generation can achieve continuous and stable power generation and other advantages. However, due to the construction cost, higher requirements for environmental factors, large-scale promotion is still difficult (Yang et al., 2017).

Biomass is also the world's fourth-largest energy source, following coal, oil, and natural gas, and is considered as one of the most promising energy and an attractive feedstock for three main reasons (Geng et al., 2020). Firstly, it is an abundant and widely distributed renewable resource that could be sustainably developed in the future. Secondly, it has positive environmentally friendly properties, such as reducing greenhouse gas, NO_x and SO_x emissions depending on the fossil fuels displaced. Thirdly, it appears to have significant economic benefits derived from the increase of fossil fuel prices in the future (Alidrisi and Demirbas, 2016). Presently, biomass meets about 10% of the total global energy demand thus making it the most potential candidate amongst all renewable energy sources (Kumabe et al., 2008). Researchers on biomass energy mainly focus on gasification (Schäfer, 1999), curing (Özdemir and Tuncel, 2000), pyrolysis (Balat et al., 2009; Siemons, 2002), biochemical (Balat, 2008), liquefaction (Toor et al., 2014), direct combustion (Williams et al., 2015) and thermochemical (Kumar et al., 2017) of biomass energy, and on of public policy and commercializing (Zhang et al., 2004), technology and equipment (Liu and Jiang, 2008), aided generation systems (Hou et al., 2011) and situation (Zhao et al., 2012) of biomass power generation in China. It has huge reserves of biomass resources in China (Zhao et al., 2013a, b). Considering the prices of fossil fuels, the gap between energy supply and

demand, energy security, environmental issues, and the reduction of greenhouse gas emissions, the use of biomass as an energy resource has become an important and interesting issue (Alcala et al., 2011; Enagi et al., 2018). According to statistics, the total installed capacity of biomass power generation has reached 14.76 million kW by 2016 in China, ranking the second in the world (China Energy Portal, 2017). A report by the World Energy Council estimates that by 2050, bioenergy will account for 25%–33% of the world's energy supply (de Almeida D'Agosto et al., 2015). In China, the utilization of crop straw was 720 million tons in 2015, and 100 million tons (11.4%) had been utilized as energy utilization. The proportion of renewable energy consumption in China will be 20% in 2030 (Zheng et al., 2018). For the use of renewable energy, the electricity generation from renewable resources will be more than 80% of the total global electricity supply by 2040 (Liu et al., 2017). Three main areas of biomass energy use should be highlighted: direct combustion to produce heat and electricity (solid fuel: charcoal or molding fuel), gasification to produce combustible gases (gas fuel: hydrogen, biomass gas, and biogas, etc.), and production of synthetic liquid fuels (biodiesel, biooil, methanol, ethanol and vegetable oil, etc.), which are burned to produce heat and electricity (Cao, 2013).

Direct combustion power generation is the direct combustion of biomass in the boiler to produce steam to drive the steam turbine and generator to generate electricity (Malinowski et al., 2012). Therefore, in the biomass raw material collection must pay attention to the biomass is not easy to coalescence and low calorific value characteristics, to ensure that it is in the compact forming state of direct combustion power generation (Montingelli et al., 2015). In 1988, the first biomass direct combustion power plant was built (Lin and Song, 2005). In terms of the raw material collection methods, power generation methods of direct combustion power generation, the current status and level of global biomass direct-fired power generation, the factors for improving the efficiency of the power generation process, the best parameter conditions and the major equipment, the researchers in China have already been explored straw direct combustion power generation (Zhang, 2007), biomass power plant development (Wang and Zhou, 2006), materials pretreatment technology (Ling et al., 2015; Xia et al., 2018), design and running (He, 2013), vibration row furnace (Jiang and Zhao, 2009; Yang, 2009), ash characteristics and potassium resource utilization (Wang et al., 2018a, b, c, d), present situation and policy suggestions (Qi et al., 2011), cost control and scheduling (Zhao et al., 2013a, b), key issues of clean development mechanism (Liu et al., 2008), and risk management (Hu and Wang, 2010). However, direct combustion power generation has many

limitations. First, the economic indicators and the sustainability has not reached the advanced level in the current situation. Second, there are many schemes of biomass direct combustion power generation. Third, technology requirements for pollutant treatment are very high (Gaurav et al., 2016).

By using pyrolysis gasification reaction at high temperature, anaerobic fermentation reaction of micro-organisms and high-temperature gasification device, biomass can be released in the form of methane and carbon monoxide to form combustible gas (Boiesen et al., 1993). It can improve the commercial value of the biomass power generation industry, and effectively reduce the operating costs of enterprises (Boiesen et al., 1993; Klerk, 2013). Then, biogas power generation technology is widely used in some power generation countries and occupies an important position in energy production (Sun et al., 2018). But biogas power generation lacks sufficient competitiveness due to the deficient resources, the high price, the high feed-in tariff and some other factors (IEA, 2019; Wang et al., 2018d).

The combustion of biomass causes environmental pollution and is not very efficient, as the full potential of the biomass is not exploited (Hasegawa et al., 2010). Biomass-derived liquid fuels are termed as biofuels, which can be produced practically in most places, and is also beneficial to global environmental protection and can even achieve zero CO₂ emissions (Hu et al., 2017; Sun et al., 2018). Biomass liquid fuels mainly include bioalcohol, biodiesel oil, biomass pyrolysis oil and so on, and their raw materials are generally livestock manure, industrial organic wastewater, plants, urban living garbage, and etc. Liquid biofuels can be used as fuel for heating and power generation (European Parliament, 2009). They have the advantages of high energy densities, convenient storage and transportation conditions, and a wide range of applications (Zhang et al., 2016b). They can also reduce the cost of desulphurization per unit power supply from 22.16 RMB/MW per hour to 16.06 RMB/MW per hour (Liao et al., 2007). Biomass liquid fuel can be directly used as fuel for various industrial oil-fired boilers, internal combustion engines (ICE) and engines through technological modifications (Song et al., 2020). For instance, compressed natural gas (Carlucci et al., 2008; Monsalve-Serrano et al., 2020), biodiesel and ethanol (Shamun et al., 2018) have been supplied and tested on internal combustion engine as alternative fuels, and it can be implemented improving the CO₂ and the NO_x-soot trade-offs using specific bowl design (Dimitrakopoulos et al., 2019), innovative fuel injection systems (Nyrenstedt et al., 2021) and biofuels. Therefore, biomass liquid fuel has flexible, adaptable, clean, and other characteristics.

The United States has more than 450 biomass power plants, and the number is still growing (Liu et al., 2014). These power plants reduce carbon dioxide emissions and increase their per capita GDP (Bilgili et al., 2017). The largest power plant in Britain had used biomass fuel after transforming coal (Kjärstad and Johnsson, 2007). Turkey's biomass fuel accounts for 14% of the country's energy production, which is equivalent to 33 million tons of oil (Toklu, 2017). As the main biomass resource in the future, there are about 30 billion litre biofuels used in the world every year (Ullah

et al., 2014). Global biomass power generation increased from 277.1 GWh in 2009 to 495.4 GWh in 2017, in which, the global installed capacity of straw power generation was 18.68 GW, accounting for 15.91%, and the installed capacity of bioliquid fuel power generation was 2.87 GW, accounting for only 2.44%. China accounts for 11.23% of the global biomass power generation capacity with 13.22 GW, ranking the second (Figs. 1–3) (Patowary and Baruah, 2019). China's average annual forestry waste and processing residues are up to hundreds of millions of tons (Malinowski et al., 2012). These resources' calorific values of general combustion are as high as 4000–4800 kcal/kg (Kumabe et al., 2008). It is imperative to develop biomass fuel in China.

A 15 MW biomass power plant can consume 203,000 tons of abandoned agricultural and forestry crops, which is equivalent to 63,500 tons of standard coal, and can reduce CO₂ emission by at least 10,000 tons per year. Then, biomass liquid fuel power generation is helpful to optimize the decarbonized power sector portfolio in light of the new sustainable normal of Chinese economy in the coming decades (Waisman, 2015). The Indian experience shows that a 10-MW biomass power project can create employment for approximately 160 people during the 18-month construction phase (Kumar et al., 2015), and help alleviate the local lack of electricity (Patowary and Baruah, 2019).

Finally, this overview is to promote the application of biomass liquid biofuel in China from policies and finance environment, power generation devices, fuel application situation and performance. Then, it is effective to create jobs, generate revenue, obtain social-economic stability, save fossil energy, reduce pollution, optimize the energy structure, and utilize the abandoned agricultural and forestry crops.

2. Policies and finance environment

Policies and the finance environment are the chief key factors affecting the foundation and development of biomass power generation, shown in Fig. 4. In European Union, based on fundamental and applied research in combination with industrial investigations on biofuels and bioliquids, the first renewable energy directive was adopted in 2008. At the same time, through the “strategic energy technology plan”, the development of biotechnology is proposed to promote the development of the low-carbon economy. In order to achieve

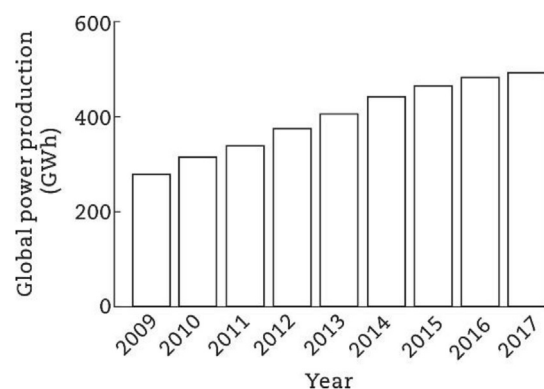


Fig. 1 – Situations of global biomass power generation.

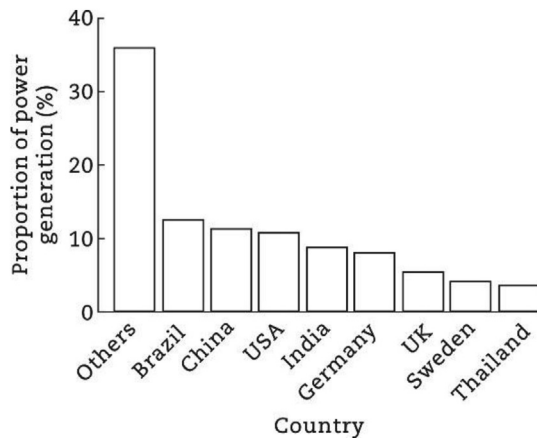


Fig. 2 – Proportion of power generation in different countries in 2017.

the goal of compulsory execution under the EU's policy framework, each member country has successively issued various policies and regulations to promote the development of the biomass energy industry, equipment and products (Balat, 2011). By the European Union Horizon 2020 PACE project, more than 500 fuel cell micro-combined heat and power systems have been deployed in Belgium and momentum is forming for the large-scale application of this innovative and efficient energy solution. The National Balancing Point in UK encourages industrial demonstration of bioethanol production using agricultural and forestry residues (cellulose) as feedstocks. Industrial-scale demonstration projects for cellulosic hydrolysis and ethanol electricity co-generation of liquid fuels and bio-based chemical products will be further developed (Flamos et al., 2011).

China has abundant biomass energy resources, which were preliminarily estimated to have a resource potential equivalent to 10 billion standard coal (Lan, 2015). The 13th five year plan (2016–2020) had emphasized the development of renewable energy. Biofuel was included in the national energy development strategy and listed as a key project in China (de Almeida D'Agosto et al., 2015; Lan, 2015). In the 13th five year plan for energy development issued by China

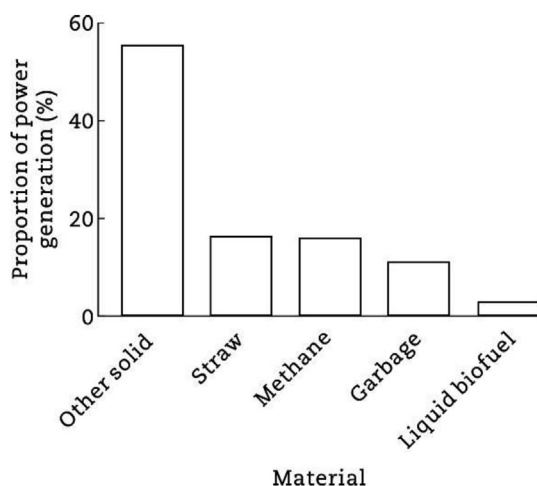


Fig. 3 – Material proportion of power generation in 2017.

in 2016, it is mentioned that we should actively develop biomass liquid fuel, gas fuel, and solid briquette fuel (Amirante et al., 2019). The Chinese government has clearly proposed to expand the production of non-grain liquid biofuel for commercialization in the provinces or regions which have rich sources of feedstocks and good processing bases (Kurji et al., 2016). The potential for rural biomass liquid fuel power generation is enormous, especially in remote areas where centralized power generation system and grid penetration become uneconomic (Wang et al., 2019). Significant results could be achieved if the biomass liquid fuel power generation industry was sustained and vigorously developed in conjunction with poverty reduction strategies (Ackom et al., 2011; Qiu et al., 2013). At present, 28 provinces (cities and districts) have developed liquid biofuel power generation projects. The total approved capacity reached 12,226.21 MW, 7790.01 MW of which from biomass, accounting for 63.72% (Zhang et al., 2017). In 2013, China's biomass power generation equipment has reached 7.5 million kW installed capacity (Zhou et al., 2016).

In 2020, the biomass energy industry has increased the investment by about 196 billion yuan. In the investments, 40 billion yuan has been used for biomass power generation, 120 billion yuan for bionatural gas, 18 billion yuan for the biomass fuel heating industry, and 18 billion yuan for bioliquid fuels (Patowary and Baruah, 2019). Local subsidies for biomass power generation have been listed in the 14th five year plan. It is reported that the state will increase the share of biomass electricity price subsidies by local governments year by year, until the central financial withdraws. Opinions regarding the deepening of the power sector's reform, released by China's State Council in March 2015, highlighted that the country should be propelled to further optimize its energy mix and improve the share of renewable energy in electricity generation (Lan, 2015). China's Industrial Biotechnology White Paper and Investment Analysis Report on China's Biological Industry 2017 pointed out that China's fuel ethanol production is expected to usher in broad space for growth. In April 2018, the National Energy Administration in Beijing has completed the preliminary formulation of the overall layout of the biofuel ethanol industry and continues to promote the application of biofuel ethanol in an orderly manner. In fact, the country, as early as 2000, began to focus on liquid biofuel power generation. The 588 policy plan in 2011 provides overall direction for biofuel (Chen et al., 2016a). It supports and promotes the research and development of key technology and specialized equipment for the production process of non-grain fuel ethanol, biodiesel, biogas, biohydrogen and other bioenergy products. The 216 policy in 2012 gives specific measures and objectives for the development of biofuel.

According to the World Energy Investment Report 2018, more than 750 billion dollars was spent in the power sector in 2017. Investment to increase renewable energy capacity (excluding hydropower) in power generation reached a record high, amounting to 280 billion dollars with a year with a yearly growth rate of 2.2% (REN 21, 2018). China alone invested more than 126 billion dollars in 2017, and the annual growth rate is 23%, in line with developing countries showing a sharp rise of 20% (REN 21, 2018). In contrast, renewable energy investment by the developed nations decreased by 19%. In 2016, globally,

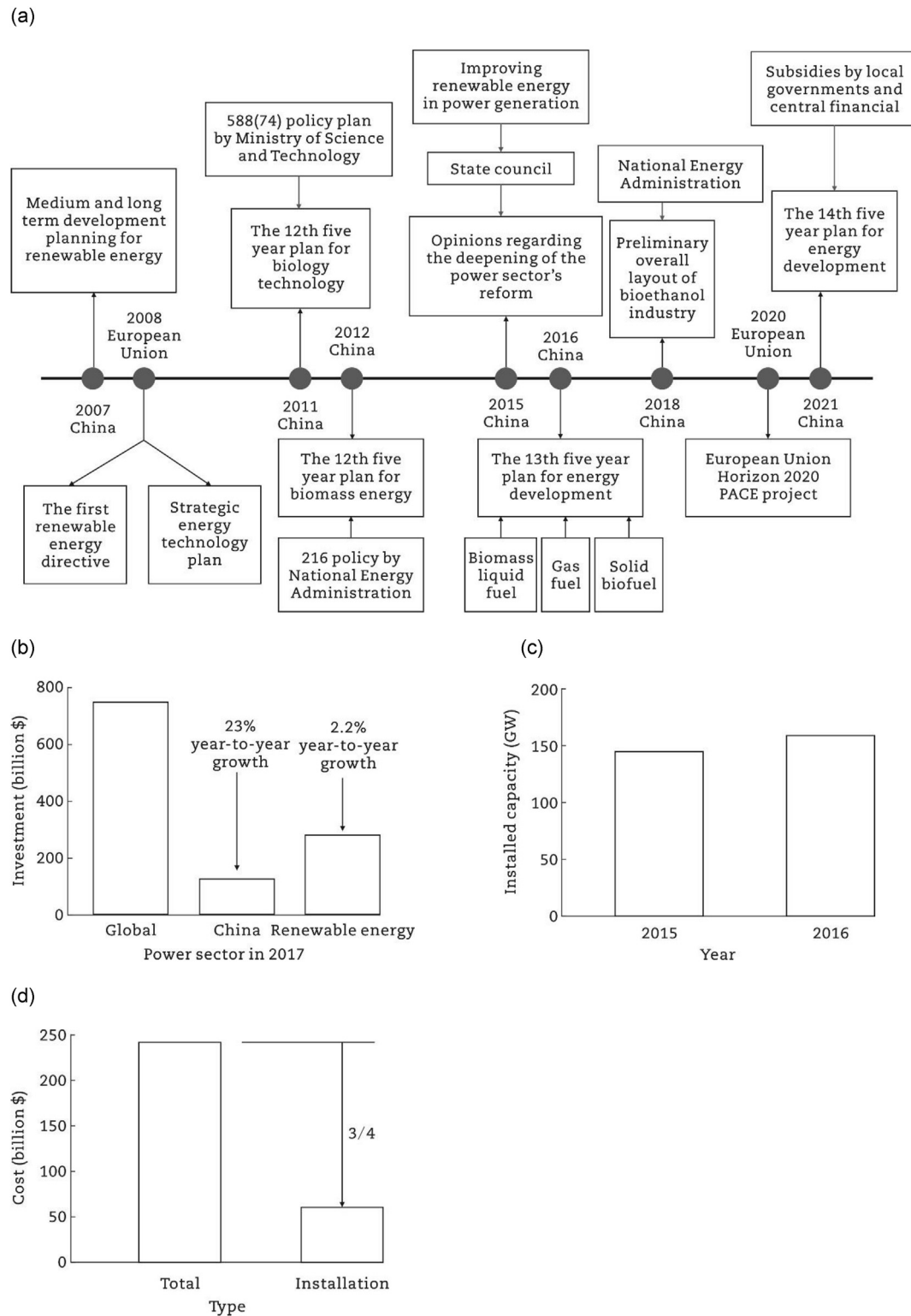


Fig. 4 – Policies and finance environment. (a) Policies of China and European Union. (b) Investment in the power sector in 2017. (c) Installed capacity. (d) Cost in 2016.

the clean energy installation was 160 GW, a 10% rise from the previous year, installed at a quarter of the expense.

In sum, with the economic development, the increased social demands, and the promotion of policies and finance, liquid biomass fuels power generation became important to

power generation as an important complement. Biofuels power generation plays an important role and is valuable in renewable energy application. As a result, the liquid biomass fuels, corresponding devices and performance became the focus in the fields of power generation.

3. The biomass liquid fuels for power generation

Under the guidance and encouragement of policies and finance, various biomass fuels were used for biomass power generation. Liquid biomass fuels have more advantages and attract more attention. At present, the mainstream liquid biomass fuels mainly include biomethanol (CH_3OH) and bioethanol ($\text{C}_2\text{H}_5\text{OH}$), bioglycerine and biodiesel, etc. (Vasudevan et al., 2005). The applied liquid biomass fuels on power generation were retrospectively analyzed as following.

3.1. The obtainment of biofuel

Biomass can be liquified to produce biooil at very high pressure or through a fast pyrolysis path (Montingelli et al., 2015). The biooil has a volumetric energy density up to 10 times higher than biomass and is more convenient for transportation (Xiu and Shahbazi, 2012). The Fourier transform infrared spectroscopy (FTIR) spectrum of biooil showed that the component of the biooil includes acids ($\text{C}=\text{O}$ 1717.2 cm^{-1}), alcohols ($\text{O}-\text{H}$ 3404.6 cm^{-1} , $\text{C}-\text{O}$ 1268.9 cm^{-1}), ketones ($\text{C}=\text{O}$ 1717.2 cm^{-1}), aldehydes ($\text{C}=\text{O}$ 1717.2 cm^{-1}), ethers ($\text{C}-\text{O}-\text{C}$ 1052.3 cm^{-1}), phenolic compounds ($-\text{H}$ 3404.6 cm^{-1} , $\text{C}-\text{O}$ 1268.9 cm^{-1}), etc. (Fig. 5) (Lyu et al., 2015).

At present, biomass-to-alcohol fuel is one of the most popular approaches and it accounts for the highest proportion (Xu et al., 2020). The first generation biofuel was seriously restricted (Naik et al., 2010). The second generation biofuel, which is obtained from cellulose (Mungodla et al., 2019), hemicellulose and lignin in crop straw (Song, 2011), cassava and waste paper (Duff and Murray, 1996; Mao et al., 2017), sawdust (Hadrich et al., 2018; Xuan and Leung, 2011), was considered to be one of the most promising biofuels (Qiu et al., 2010). The prepared technologies include step-by-step hydrolysis fermentation, synchronous saccharification fermentation, synchronous saccharification co-fermentation, non-isothermal synchronous saccharification fermentation, direct microbial transformation, etc. (Öhgren et al., 2007; Saha et al., 2005; Shrivastava et al., 2014). The enzymatic hydrolysis process has the characteristics of mild

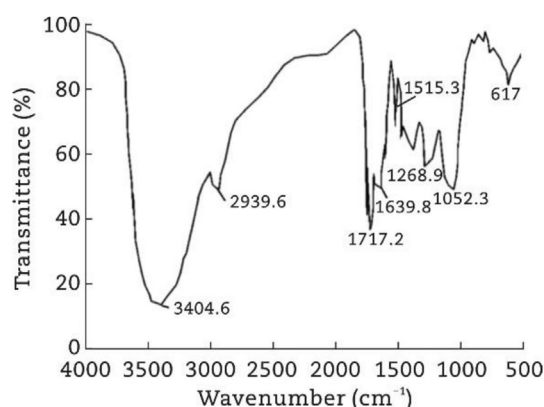


Fig. 5 – FTIR spectrum of biooil.

conditions, low energy consumption, no toxic degradation products, high conversion rate, no corrosion, no pollution and no fermentation inhibitors. But the amount of cellulase per unit is large, which leads to the increase in cost (Jaroenphasemmesuk and Tippayawong, 2015; Lu et al., 2011). Biomass gasification synthesis of liquid fuel has attracted much attention due to its wide adaptability, diversification of products and high cleanliness. But the selectivity and yield of ethanol are very low (less than 20%) (Wang et al., 2018c). Biomethanol, bioethanol and biobutanol are the typical second generation biofuels (Malinowski et al., 2012), and the biomethanol is likely to be more favorable due to its better CO_2 mitigation potential (Hasegawa et al., 2010). Solid oxide electrolysis cell can achieve higher efficiency on biomethanol production (Zhang et al., 2019). After optimizing the kind of catalyst, the hydrolysis temperature and fermentation temperature, the conversion rate of bioethanol can be up to 90% or more (Brethauer and Wyman, 2010; Zhu et al., 2006). Bioethanol can be also produced from kelp residue and the conversion rate can reach 85.16% (Jin et al., 2015).

In addition, biodiesel can be hydrolyzed on a large scale from microalgae rather than other raw materials (Hadrich et al., 2018). Ethylene glycol (EG) can be produced from biomass raw materials through hydrolysis, direct fermentation, direct catalytic reaction, or catalytic hydrogenation (Crabtree et al., 2006). Also, hydrocarbon fuel can be produced from biomass through pyrolysis, liquefaction and hydrofining (Yang et al., 2020; Zhang et al., 2014b).

3.2. The application of biofuel on power generation

Methanol was used to substitute petroleum oil and natural gas for gas turbine power generation due to its higher energy efficiency (Azadi et al., 2016). Direct methanol fuel cell (DMFC) is an important branch of fuel cell (FC), and the catalyst plays an important role in energy efficiency. The Pt-Sn/C catalyst is an effective solution of poisoning resistance in DMFC compared to traditional catalyst (Gong et al., 2018). Stability is an important guarantee for energy efficiency of DMFC and methanol cracking catalyst is the critical factor (Lattner and Harold, 2007). In addition, reaction temperature, contact time, and the amount of mixed air are also important parameters for improving energy efficiency of DMFC (Agrell et al., 2002; Caglar et al., 2015). Besides high energy efficiency, methanol attracted much interest by self-reforming to become carbon neutral. In order to improve the efficiency of self-reforming, Pt-based regulative accelerators and granular $\text{Cu}/\text{ZnO}/\text{Al}_2\text{O}_3$ accelerators have been used to increase the conversion efficiency of hydrogen production substance to 90% (Dalena et al., 2018). The membrane transfer technique with high temperature is used to improve CO resistance (Fukuhara et al., 2005; Maximini et al., 2012). The selective oxidation technology had been used to remove CO from the reconditioning gas (Moreno and Wilhite, 2010). Pd-Ag membrane metal tube in a packed bed reactor can increase 5%–10% conversion rate of methanol (Chen et al., 2009). Thus, methanol has very high applicative valuation in FC due to high energy efficiency in FCs.

Bioethanol has also been applied in the field of biofuel power generation in FC. Its energy efficiency is dependent on catalyst material and operating conditions (Steil et al., 2017). The energy efficiency of ethanol FC can be promoted by carbon borne Pt catalysts (Rousseau et al., 2006; Zhou et al., 2004) and electrolyte membranes (Fu et al., 2007; Maab and Nunes, 2010; Yang et al., 2008), which were favorable to realize higher catalytic oxidation activity (Wang et al., 2008). Even so, ethanol is not a better choice in FC power generation compared to methanol.

Biodiesel production has increased from 7.2 billion litres to 41 billion litres in the past 15 years, and global carbon dioxide emissions are estimated to decrease by 25% by 2050 based on the utilization of biodiesel (Barua et al., 2020). The potential reduction of the CO₂ emission is 1057 ktons when 10% of petroleum diesel is replaced by biodiesel. This shows that biodiesel has the potential of approximately 8% CO₂ reduction when it is used in the power sector compared to the transportation sector (Cao, 2013). The application of biodiesel presents a great impact on CO₂ reduction for the power sector. Accordingly, it is more significant to use biodiesel in a microturbine than in an internal combustion engine (Chisti, 2007; SathyaSelvabala et al., 2011; Somorin et al., 2017). Direct use of biodiesel will also lead to the deactivation of carbon deposition of the battery anode. At present, the research on the application of biodiesel in solid oxide fuel cell (SOFC) power generation mainly focuses on the field of external reforming, through which the biodiesel is transformed into small molecular fuel for application. Or a certain amount of alcohol, such as methanol and ethanol, is usually added to settle catalyst carbon deposition (Tartakovsky and Sheintuch, 2018).

The direct ethylene glycol (EG) fuel cell prepared by proton exchange membrane has the characteristics of low fuel permeability and high energy density (Peled et al., 2002). Specifically, the specific energy of EG is 5.27 kWh/kg, and the theoretical efficiency of the cell is 98.9% (de Lima et al., 2003). Some studies have shown that EG is a better choice for alkaline direct oxidation fuel cells (ADOFCs) than ethanol (An et al., 2010; de Lima et al., 2003; Matsuoka et al., 2005). EG has been recently used to ADOFC with alkali-doped polybenzimidazole (PBI) membranes technology, and also can yield a peak power density of as high as 112 MW per cm² at 90 °C (Hou et al., 2008a, b). The performance of both electrodes of a nanoporous proton-conducting membrane-based acid direct fuel cell can be significantly improved after using EG (Livshits and Peled, 2006). However, in the process of oxidation, EG produces an intermediate that can be adsorbed on the single metal catalyst, and then weakens the performance of the catalyst (Motoo, 1975). The preparation of binary or multivariate catalysts can improve the performance of EG fuel cell (Neto et al., 2005). Moreover, EG was also used as an alternative fuel for gas turbine and internal combustion engine (ICE) power generation (Cormos et al., 2015).

Whether in terms of obtainment or application or physical and chemical characteristics or carbon-neutral, methanol, ethanol, ethylene glycol and biodiesel show strong application potential in power generation. Comparatively speaking, methanol is the best one for widely used in various devices, fuel reforming from CO₂ and CO₂ reduction.

4. Power generation device of biomass liquid fuels

4.1. Individual power generation system

Based on the above introduction of liquid biofuels for biomass power generation, it is necessary to select suitable and effective power generation devices. Power generation devices with bioliquids is classified into ICEs, turbine engines (often denoted as gas turbines) and advanced power generation technologies (Seljak et al., 2020). The power generation efficiency of a 20 MW single-cycle gas turbine is only 30%–35%. ICE with lean burn technology is more efficient and the efficiency is up to 44%. It means that under the same output power, fuel cost of gas turbine is 9%–14% higher than that of ICE (Liu, 2002). Thus, in the low-power market, the market with a capacity of 5×10^4 kW or less, the ICE is gradually becoming a competitive product (Liu, 2002). The system of turbines has a recuperator that recuperates hot exhaust air to be channeled back into the intake, thus enabling the users to install other combined heat and power (CHP) applications (Dimitriou et al., 2018; Singhyadav et al., 2012). As a result, they have higher efficiencies than ICE (Islam et al., 2014).

FC is a kind of installation that can directly convert chemical energy into electrical energy, and its power generation efficiency may be up to 83% (Jiao et al., 2011). The reaction product of FC is only water, because the reaction raw material is gas and does not go through the over-high temperature combustion process, and the tail gas almost does not contain dust, nitride and trioxide (Ahmed and Krumpelt, 2001). Therefore, FC is a highly efficient and environmentally sustainable energy source device. The fuel cell converts the chemical energy of biofuel into electrical energy and heat through electrochemical reactions. It is a highly efficient energy conversion device, which conversion efficiency is more than 60%, and much higher than that of fossil energy in internal combustion engines or in thermal power plants (Iqbal and Wang, 2006). Micro-biofuel cell used microorganisms as a catalyst to oxidize organic and inorganic substances and generate electric energy, and was rarely studied in the field of co-generation with other systems due to low power density (Rahimnejad et al., 2015). In view of renewability of raw materials and carbon neutrality, many biofuel FCs had been developed and applied for direct utilization, internal reforming or external reforming of biofuel (Damle, 2008; Soloveichik, 2014). Fuel cells include molten carbonate fuel cell (MCFC), SOFC, phosphoric acid fuel cell (PAFC), alkaline fuel cell (AFC) and proton exchange membrane fuel cell (PEMFC) (Damle, 2008). The utilization of liquid biomass fuel in the fuel cell can be divided into three ways: direct utilization, internal reforming and external reforming. Direct utilization means that biofuels directly participate in electrochemical reaction with hydrocarbon gas molecules instead of transforming into H₂, CO and other small molecular fuels. The inner reforming is the in-situ reforming of the fuel in the electrode layer. External reforming is the integration of the catalyst into an external unit to be used as a reforming component of the

SOFC system (Soloveichik, 2014). MCFC is the largest single-capacity fuel cell and the most likely commercial high-temperature fuel cell (Ong et al., 2017). In addition to the distributed power generation, MCFC can also be combined with integrated coal gasification combined cycle (IGCC) power generation to form a more efficient integrated coal gasification fuel cell system, the coal-based power generation efficiency is above 55%, and can achieve near-zero emissions simultaneously (Soloveichik, 2014). Liquid biomass SOFC power generation system has an excellent application prospect in the fields of transportation, vehicle power, special fields, stationary/distributed power stations, urban/household cogeneration, auxiliary power units, mobile/portable power sources and so on (Zhang et al., 2016a, b). SOFCs have a higher current and power density than first-generation and second-generation fuel cells (represented by PAFC and MCFC). Moreover, SOFC can better achieve cogeneration (Chen et al., 2016b; Ha and Wachsmann, 2019), and the comprehensive energy utilization rate is up to 80% (Cardoso et al., 2018; Chino et al., 2020), combined with compression type capture absorption-type heat pump technology (Chino et al., 2019). PAFC can use air as cathode reaction gas or catalytic reforming gas as fuel, which makes them widely used in fixed power stations (Ong et al., 2017). AFC must use pure oxygen as an oxidant, which directly increases the cost of commercial applications. Accordingly, the present commercial application of AFC is not popularized, and it is only used in some special military fields (Velisala et al., 2015). PEMFC has the characteristics of low operation temperature, high energy density, clean and pollution-free (Nguyen and Shabani, 2020). Direct methanol fuel cell (DMFC) is selected as PEMFC, which has the characteristics of low-temperature quick start, clean fuel and environmental protection, and simple battery structure (Gao et al., 2020; Yang, 2019). Its efficiency is about 40% (Yu, 2020), and its working principle is shown in Fig. 6. DMFC may become the mainstream of future portable electronic product applications.

Alkaline conditions can be described as follows.

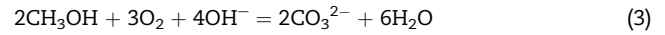
At the negative electrode:



At the positive electrode:

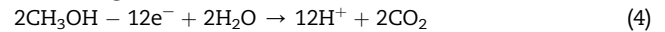


The total reaction formula:



Acidic conditions can be described as follows.

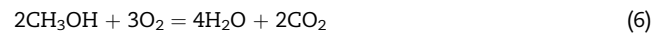
At the negative electrode:



At the positive electrode:



The total reaction formula:



4.2. Combined power generation system

4.2.1. Gas-steam combined cycle

The gas-steam combined cycle is a thermal cycle that combines gas turbine cycle with steam turbine cycle. Through cascade utilization of energy, the efficiency of the combined cycle generator set has reached more than 57% (Tan et al., 2014).

4.2.2. The combined power generation of biomass and solar Trough solar power generation system is more suitable for combined application with biomass power generation system. Biomass energy solar energy combined power generation device can be divided into three parts: solar collector, circulating fluidized bed boiler and steam turbine power generation system (Pantaleo et al., 2020). Biomass energy power generation

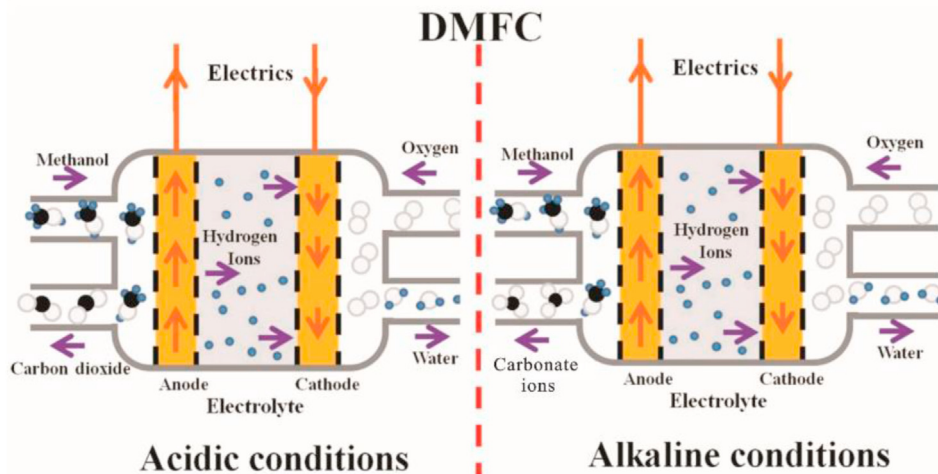


Fig. 6 – Structure and principle of DMFC.

system is only different in boiler equipment and fuel type to conventional thermal power generation system, and steam turbine and generator at the back end of photo-thermal power generation are compatible with thermal power generation (Bhattacharya et al., 2011). The main obstacle for commercial application is still that the cost is higher than that of conventional thermal power (Firmansyah et al., 2018).

4.2.3. CHP

Liquid biomass fuel can be directly used in industrial boiler combustion and direct combustion of various industrial furnaces for power generation, and the exhaust temperature and unused fuel gas can be combined with a gas turbine or steam turbine to combined cycle power generation system (Alidrisi and Demirbas, 2016). The use of absorption heat pumps to recover waste heat from biomass direct combustion flue gas has considerable benefits of energy saving and pollution reduction (Xu and Wang, 2017). CHP systems can convert the energy of biofuels into power and heat simultaneously with high efficiency (Seljak et al., 2020). The main forms of CHP are boiler-heated steam turbine CHP system, gas turbine CHP system, internal combustion CHP system and fuel cell

system (Vialeto et al., 2017), and the energy utilization rate may be higher than 80% (Xie et al., 2012). The heat in CHP can be used for space conditioning, industrial processes or as an energy component (Demirbas, 2007). Compared with 35%–37% overall efficiency of traditional fossil fuel power generation systems, the overall efficiency of general heat and power systems can reach 60%–80%, and the most advanced combined heat and power systems can reach over 90% (Wu and Wang, 2006). CHP systems mainly include coal, gas, waste incineration CHP systems, biomass CHP systems, solar CHP systems, and other CHP systems (Oland, 2004). Compared with other combined heat and power technologies, fuel cell combined heat and power (FC-CHP) systems have higher power generation efficiency and comprehensive energy utilization, and are unparalleled in terms of economy, environmental protection and reliability (Olabi et al., 2020; Otomo, 2016). Comparing the energy and exergy of each reactor in FC-CHP system in Fig. 7, DMFC has the largest energy loss and the lowest energy efficiency and accounts for more than half of the overall exergy loss of the system (Sui et al., 2019). Therefore, the fuel cell should be improved in alkaline anion exchange membrane (Pan et al.,

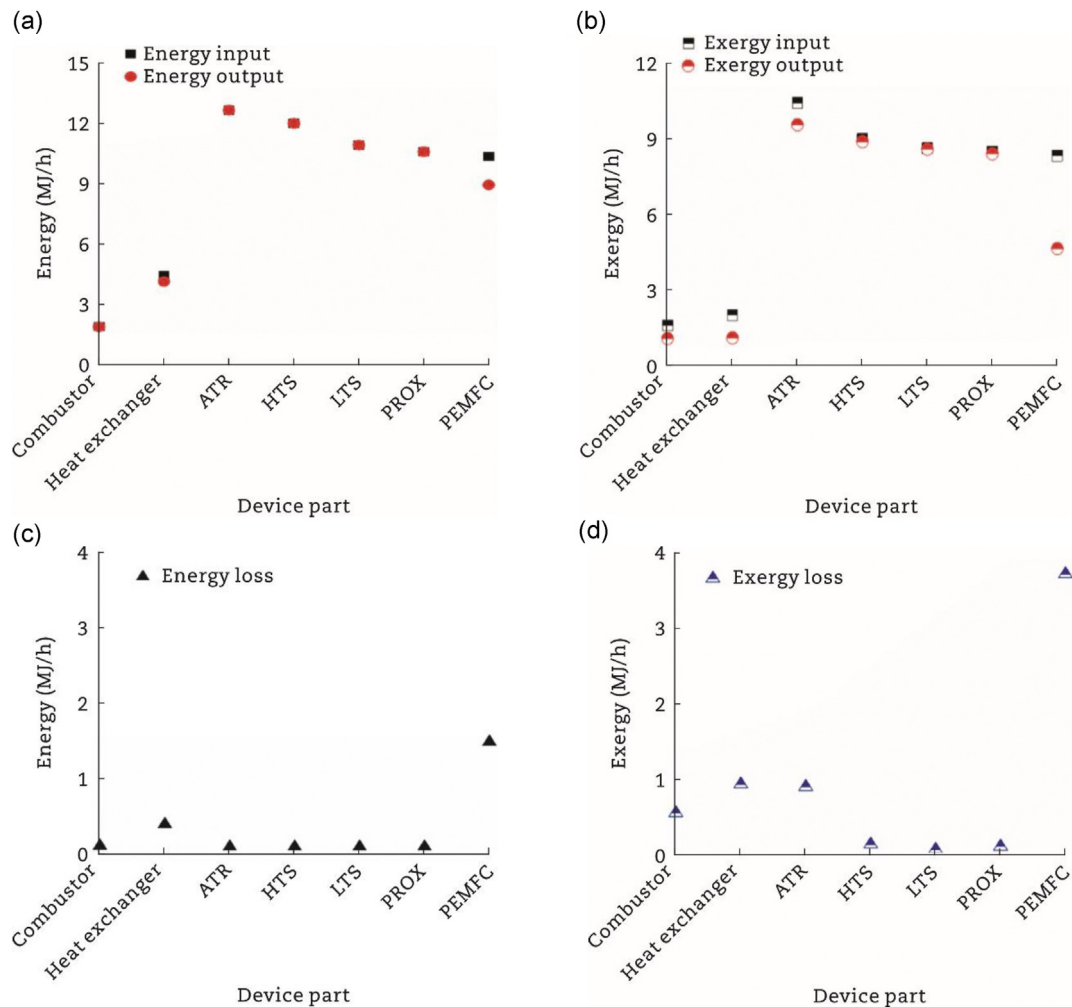


Fig. 7 – Energy and exergy analysis of FC-CHP system. (a) Energy input/output. (b) Exergy input/output. (c) Energy loss. (d) Exergy loss.

2017; Sui et al., 2019), catalytic (Liu et al., 2018), cost (Löbberding and Madlener, 2017), economics (Wang et al., 2018a, b, c, d), energy and exergy loss and the operating temperature (Luo et al., 2016). Then, the exergy loss of the FC-CHP system will be effectively reduced.

The SOFC + gas turbine (GT) combined cycle can improve the efficiency of power generation to 55%. SOFC + GT + steam turbine (ST) three-stage combined cycle had a combined thermoelectric utilization efficiency of 73%. The SOFC + GT + ST three-stage combined cycle power generation system is being tested with an efficiency of up to 65% (Jurado and Valverde, 2003). There are three biomass integrated coal gasification combined cycle (BIGCC) power generation demonstration projects with an installed capacity of 6–10 MW in the UK and the US (Alcala et al., 2011). The power generation efficiency can reach 40%. Integrated coal gasification fuel cell (IGFC) is the combination of IGCC and SOFC power systems, improving the efficiency to 50% or more and reducing water consumption to 80% of the IGCC (Stambouli and Traversa, 2002). The combination of liquid biofuel with IGFC technology can achieve clean, efficient and quiet power generation, effectively alleviate environmental pressure and reduce ultra-carbon emissions. Consequently, IGFC is an important direction for “green coal power” and is one of the leading technologies for biomass energy conversion (Jansen et al., 1994; Vasileiadis and Ziaka-Vasileiadou, 2004). The above cutting-edge technologies such as a biomass fuel integrated electricity and heat generation system (Fantozzi and Bartocci, 2017), biomass gasification for combined heat, cool and power generation (Wang et al., 2016), biomass co-combustion and/or the biomass methanol fuel cell power generation pathway (Carter et al., 2013; Naik et al., 2010) offer large potentials for deeply decarbonizing power sector.

Finally, the application of biomass renewable energy on electric power generation is shown in Fig. 8.

5. The performance of biofuels for power generation

Targeted at biomass direct combustion power generation, the exergy efficiency is 40.36%, and the exergy loss is mainly distributed in the exhaust gas loss (Xu and Wang, 2017). The recovery and utilization of flue gas waste heat are important methods to improve the energy efficiency of the biomass cogeneration system, and also have environmental protection benefits (Zhang et al., 2020). The efficiency can be further improved by additional heat utilization, such as the production of hot water (Schneider et al., 2020). Still, biofuel is better than biomass direct combustion for power generation. The performance of power generation devices with an application of liquid biofuels determines the promotion and large-scale use of biofuel power generation. As described above, FC and ICE are the main mobile power generation devices for biofuels application, and the performance of GT or ICE is dependent on combustion of biofuels. The following parts will mainly describe the performance of DMFC and direct ethylene glycol fuel cell

(DEGFC), and the performance of ICE applied with various biofuels.

5.1. The performance of DMFC and DEGFC

As brief performance introductions of DMFC and DEGFC were summarized in section 3.2, methanol and EG both have greater potential application for FC power generation than other biofuels, it will make some complementary contents as following. As described previously, catalyst h-BN-Pt/Al₂O₃ can help to improve the DMFC energy efficiency with as high as a reaction conversion rate of 97% (Wang et al., 2010). While TiO₂ was added in the Pt-Ru catalyst, the first effect rate of DMFC raised to near 100%, and the current density increased by 5.6% (Ahmed and Krumpelt, 2001). DEGFC has a 320 mW/cm² maximum power density at 0.32 V at 130 °C, and thus EG has a theoretical capacity 17% higher than that of methanol in terms of Ah/mL (4.8 and 4, respectively), which is especially important for portable electronic devices (An and Chen, 2016). In addition, compared with glucose, sorbitol, glycerol and methanol, EG can produce the highest concentration of H₂, and release the least by-products (CH₄, C₂H₆, C₃H₈, etc.) (Li, 2016). It is particularly useful for reforming of EG. And catalytic reforming of EG can effectively inhibit the anodic carbon deactivation of SOFC, and improve the power generation efficiency and extend the service life of SOFC (Zerbinati, 2002). Accordingly, DEGFC has better performance and potential than DMFC. In fact, low yield, high price and hydrophilia limit its application.

5.2. The performance of ICE power generation with various biofuels

It is significant to develop methanol as a fuel for ICE power generation due to the effective reduction of harmful substances (Chao et al., 2000). Because of the low viscosity, good fluidity, better atomization and heat generation, methanol can achieve complete combustion with satisfying fuel-air mixing. Furthermore, methanol has obvious advantages in low combustion loss, high thermal efficiency and low emissions, although the cost is slightly increased compared with coal (Liu and Jiang, 2008). Methanol is also used for GT power generation due to the abundant sources (Li and Wang, 2004), increased thermal efficiency (Zhu et al., 2013), improved dynamic performance (Hong, 2004), good economy, and low emission (Carapellucci et al., 1996; Guiler, 2000). However, its corrosion, swelling, low ignition point and high latent heat of vaporization need to be further improved for ICE power generation.

For ICEs power generation, the addition of ethanol decreased the in-cylinder temperature, the diesel-air mixing became insufficient, and CO was the main combustion product of ethanol, so CO emissions increased with the upregulation of ethanol concentration at low and medium loads. With the augment of load, the in-cylinder temperature rose rapidly, and CO was oxidized to CO₂, then CO emissions decreased significantly, and reduced about 29%–49% along with adding ethanol at full load (Schuetzle et al., 2002). Total hydrocarbons (THC) emissions increased dramatically with adding ethanol at various engine conditions due to leaner

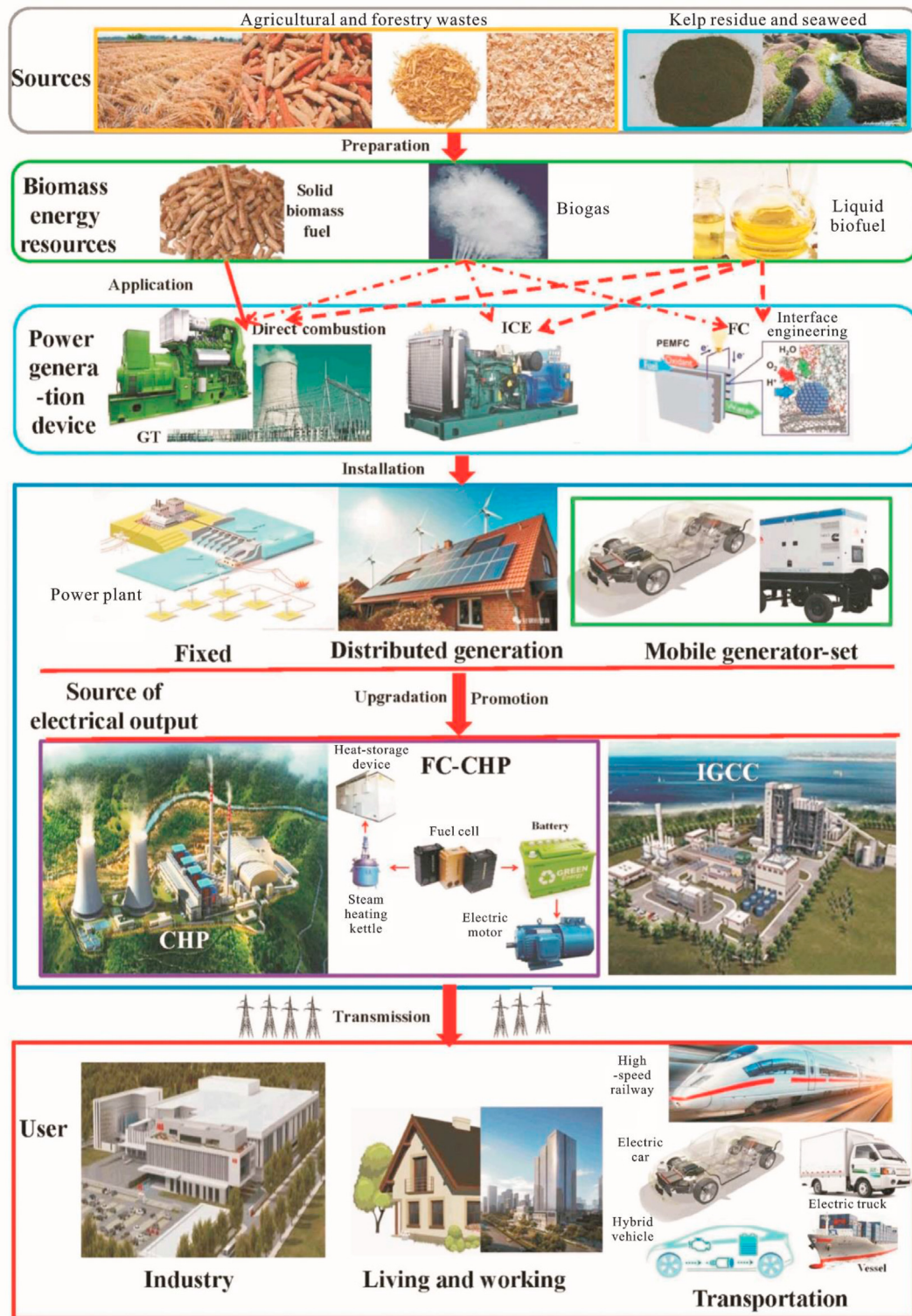


Fig. 8 – The application of biomass renewable energy on electric power generation from sources to users.

combustion, which increased the amount of unburned fuel in the quench layer and crevices of the combustion chamber (Lyu et al., 2003; Rice et al., 1991). From the view of emissions, ethanol is not a good choice for ICE power generation.

Compared with methanol and ethanol, in ICE power generation, isopropanol has better fuel economy due to higher low heating value (Esarte et al., 2012; Liu et al., 2015). Although isopropanol/diesel blends have similar combustion and emission rules on the diesel generator with ethanol/diesel

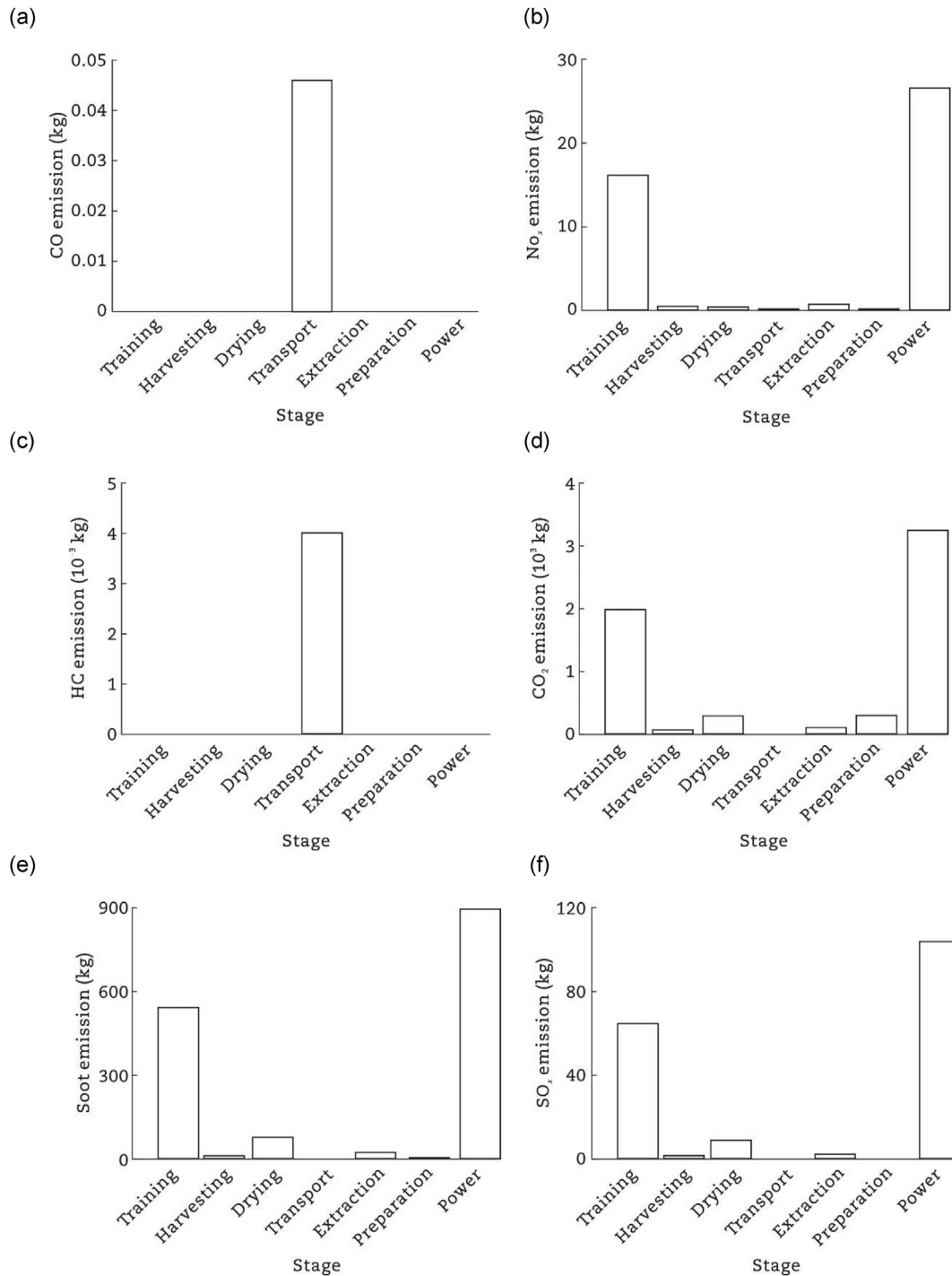


Fig. 9 – Emissions of microalgae biodiesel power system at each stage. (a) CO emission. (b) NO_x emission. (c) HC emission. (d) CO₂ emission. (e) Soot emission. (f) SO_x emission.

(Alptekin, 2017), they have higher combustion efficiency and lower CO emission and lower THC emission than ethanol blends due to higher cetane number and low heating value, and lower latent heat of vaporization (Chockalingam and Ganapathy, 2012). Meanwhile, slower heat release and lower combustion temperature lead to NO_x emission (Teoh et al., 2015). Compared with butanol and pentanol, isopropanol has higher oxygen content, which was conducive to the effective reduction of soot, PM, HC and CO emissions (Liu et al., 2015).

According to the above descriptions, isopropanol seemed to be appropriate biofuel for ICE power generation.

Compared to methanol, ethanol and isopropanol, biooil and biodiesel can be directly prepared from biomass without separation and purification. Then it has potential application in ICE power generation due to simple course of production and low cost. As mentioned in section 3.1, the alcohols, aldehydes and ethers in biooil give rise to good atomization, stable combustion and high absolute temperature, which are

beneficial to ICE power generation (Zhang et al., 2009). Owing to higher oxygen content, HC, CO and soot emissions of biodiesel generally decreased, and the reduction is also observed in the CO₂ emissions and exhaust gas temperature (Atta et al., 2020). It is shown that the CO, CO₂ and toxic organic matter emissions of biodiesel are only 10% of diesel (Dinesh and Murugavelh, 2018). Thus the potential CO₂ emission reduction for 10% diesel replacement by biodiesel was 1057 ktons for the power sector, and jatropha biodiesel power plants can reduce CO₂ emissions by 22%–76% due to the low carbon to hydrocarbons ratio and its regenerated sources (Özener et al., 2014). Biodiesel can also greatly improve the combustion and has more flame stability trends owing to the oxygen in biodiesel (Naveenkumar and Baskar, 2021). Due to lower low heating value, the break specific fuel consumption (BSFC) of biodiesel was higher than that of diesel, and the decrements were observed in engine torque, engine power, in-cylinder pressures, brake thermal efficiency (BTE) and NO_x emissions (Atta et al., 2020). Compared with neem biodiesel, mustard biodiesel has better performance in ICE power generation, which BSFC decreased slightly (0.135 kJ/kWh to 0.045 kJ/kWh) and had an upward trend of BTE due to higher oxygen content (Zheljazzkov et al.,

2012). In light of the above advantages, it has built an ICE power generator for providing ecological electricity to the public grid in Tirol, Austria, which adapted a waste cooking oil fuelled marine diesel engine and supplied 6 million kWh power every year (Zhang et al., 2014a).

With the development of production technology and huge sources of microalgae, microalgae biodiesel is a potential biodiesel for ICE power generation. Although in the process of generating electricity, CO₂ and NO_x emissions of microalgae biodiesel ICE power generation were slightly higher than that of coal-fired power due to more complete combustion (Figs. 9 and 10), microalgae biodiesel ICE power generating plants can achieve the near zero emissions of CO₂ in life cycle emission respect due to the massive CO₂ sinking by microalgae in culture ponds near the power plant (Serna-Gonzalez et al., 2014). Thus, microalgae biodiesel ICE power generation have important significance in energy conservation and emissions reduction within the scope of the controllable (Togarcheti et al., 2017).

It is more significant to use biodiesel in a microturbine as compared to ICE for mitigating the intensive effect of greenhouse gas (GHG) that occurs in the power generation sector (Nascimento et al., 2008). It seems that biodiesel in GT power generation has the significance of a more practical

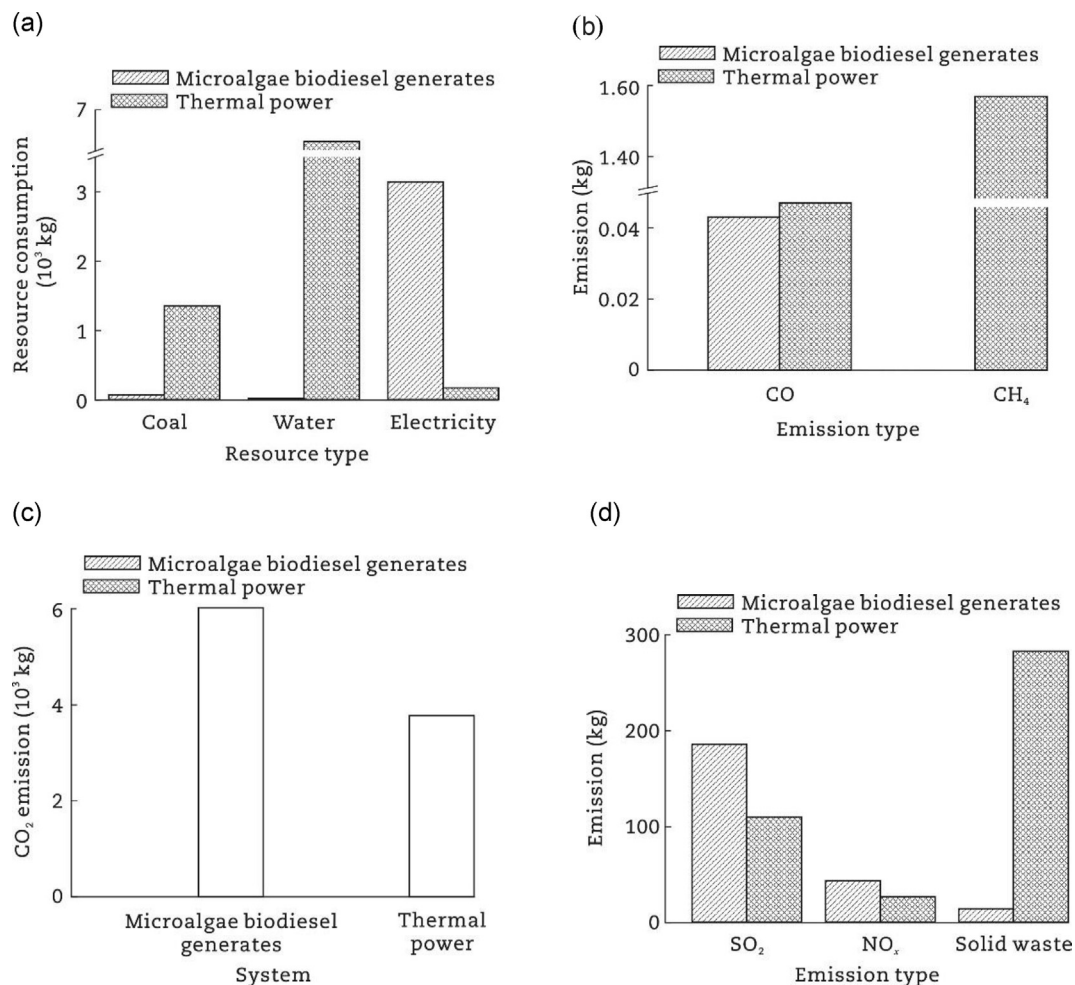


Fig. 10 – The consumption and emission of microalgae biodiesel power system and thermal power system. (a) Two system resource consumption. (b) Emission of CO and CH₄. (c) Emission of CO₂. (d) Emission of SO₂, NO_x and solid waste.

application than in ICE power generation. In reality using of biodiesel in GT power generation was limited for the increased cost of the energy generated due to the higher production cost than diesel (Kumar et al., 2013; Silitonga et al., 2013).

6. Conclusions

In this paper, biofuel power generation on policies and finance environment, applied biofuels, device and performance had been reviewed.

Despite some problems, which exist in technology R&D, policy standards, industrial system, the instability of the raw material property, the seasonal factors, the cost of storage and so on, limits biofuels power generation industrialization, biofuels have a unique potential to serve the main energy demand sectors of heat and electricity. At the same time, the development and utilization of biomass energy on biofuel power generation have become imperative in renewable and clean energy terms. Hence this paper points out investments in renewable energy power generation are increasing year by year, especially in China. Meanwhile, more policies were gradually established to support biofuel power generation. The policy guidance and investments should give strong support to scientific research to break through the key technology and achieve lower costs, then to improve product quality, economic benefits and market competitive goals at the national level. And related departments should also establish and improve the management system of renewable resources and rationally utilize resources for sustainable development of renewable resources. Also, the relevant industry should improve the standards and norms, and cultivate the industrial chain for biofuel power generation. Biofuel power generation is of great significance in the recycling of waste resources or the transformation of energy structure and environmental improvement and protection from points of energy-saving and pollution reduction, optimizing the energy structure, and utilizing abandoned agricultural and forestry crops, and also is of great significance to creating jobs, generating revenue and maintaining social-economic stability. Finally, in order to realize the goal of sustainable development, harmonious development of energy, economy and environment will also be actualized.

After review, the future development of a biofuel power generation is high-efficiency, large capacity, low-cost, portable, flexible control and combination. In stationary generating plants, GT, ICE and FC are all applicable in biofuels power generation. And CHP and IGCC are the trend in future for biofuels power generation with higher energy efficiency. By comparison, methanol, isopropanol and biodiesel are the respective best choice for FC, ICE and GT power generation. And FC-CHP, FC-ICE-CHP, FC-GT-CHP and ICE-CHP are better choices in mobile energy supply side for transport in the future. It is of great significance to energy-saving and pollution reduction and optimizing the energy structure in the traffic field.

As the development of biofuel power generation has many points of vital significance, there are a lot of work on biofuel power generation that needs to be done. This review paper

focuses on transportation energy and proposes choosing and using CHP based on the various generating facilities as an energy supplement for transport in the future after reviewing and analyzing.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

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Abbreviation

ICE	Internal combustion engine
GT	Gas turbine
FC	Fuel cell
CHP	Combined heat and power
IGCC	Integrated gasification combined cycle
MCFC	Molten carbonate fuel cell
SOFC	Solid oxide fuel cell
PAFC	Phosphoric acid fuel cell
AFC	Alkaline fuel cell
PEMFC	Proton exchange membrane fuel cell
FC-CHP	Fuel cell combined heat and power
EG	Ethylene glycol
BTE	Brake thermal efficiency
ADOFC	Alkaline direct oxidation fuel cell
BSFC	Brake specific fuel consumption
ST	Steam turbine

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